

PRODUCT REQUIREMENTS DOCUMENT

World Cup Intelligence Platform

A live, multi-tournament football analytics product

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Portfolio artifact — based on a real, production-grade system designed and built by the author.

Contents

- 1. Overview
- 2. Problem & Opportunity
- 3. Goals & Success Metrics
- 4. Non-Goals
- 5. Target Users
- 6. Functional Requirements
- 7. User Experience Principles
- 8. Release Plan
- 9. Risks & Mitigations
- 10. Open Questions & Future Work
- Appendix A — Glossary

1. Overview

The **World Cup Intelligence Platform** is a web application that lets anyone explore professional-grade football analytics for the 2026 FIFA World Cup as it happens, and for every major tournament before it. It fuses three things that normally live in separate, expensive products: a live data feed (in the spirit of FotMob), a predictive model (FiveThirtyEight-style forecasts), and a dense, interactive analyst surface (a Bloomberg-terminal feel).

The core insight driving the product: casual fans and serious analysts alike want to **play with the numbers themselves** while the tournament is live — not read a finished article after the fact. Every page is interactive, every projection is recomputed from the live feed, and the same analytical lens can be pointed at the 2026 tournament or any historical edition for context.

TL;DR A single product where a user can watch a live match, see the model's win probability update, trace a player back to their club, compare any two squads, and read an auto-generated scouting report — all from real data, across eras.

2. Problem & Opportunity

2.1 The problem

- Live football data is fragmented across apps that each do one thing — scores here, xG there, predictions somewhere else, club/transfer context in a fourth place.
- Predictive models are usually opaque: a number is presented with no way to interrogate the path behind it (how often does this team actually reach the semi-final across simulations?).
- Historical tournaments are treated as static trivia, not as a comparable analytical dataset you can switch into and explore with the same tools.
- Responsible, model-vs-market betting analysis is either absent or replaced by irresponsible "tipping".

2.2 The opportunity

Open, high-quality data now exists for both halves of the problem: **StatsBomb** publishes full event-level open data for historical World Cups, and **API-Football** provides an affordable live feed for 2026. Combining them under one normalized model makes it possible to ship a product that is simultaneously **live, historical, and explainable** — a combination no single mainstream app offers.

3. Goals & Success Metrics

The product is considered successful if it delivers a genuinely interactive analytics experience that holds up against real data, in real time, without a backend team.

Goal	Metric	Target
Live coverage	All competing teams with complete	100% of

Goal	Metric	Target
	squad, coach, standing & forecast	teams
Forecast credibility	Every probability traceable to a stated method (Monte Carlo n)	8,000 simulations / refresh
Breadth	Tournaments explorable with identical tooling	6 (live + 5 historical)
Interactivity	Pages that recompute from live state rather than static copy	Every analytical page
Reliability	Pages that render without error under partial / mid-load data	100% (graceful degradation)
Time-to-first-value	From landing to a meaningful, personalized view	< 30 seconds

4. Non-Goals

Explicitly out of scope for v1, to keep the build focused:

- **Native mobile apps.** The product is a responsive web app; native wrappers are a later consideration.
- **User accounts & social.** Personalization (favorites, custom cards) is device-local; there is no login, feed, or comments.
- **Real-money betting.** The betting surface is an analytical, model-vs-market comparison tool with prominent responsible-gambling framing — it never places or brokers a wager.
- **Proprietary data licensing.** v1 uses open and low-cost commercial feeds; premium tracking data (e.g. broadcast tracking) is out of scope.

5. Target Users

Persona	What they want	Key features for them
The Analytics Fan	To go beyond the scoreline — xG, momentum, who actually played well.	Live match center, shot maps, power ratings, player percentiles
The Forecaster	To see who the model favors and how confident it is.	Predictions, bracket, championship probability, golden boot race
The Storyteller	Narratives — breakout players, debutant nations, club-to-country threads.	Storylines, Discoveries, Club Connections, the 3D globe

Persona	What they want	Key features for them
The Edge-Seeker	To compare the model to the betting market — responsibly.	Betting Edge (EV, de-vig, fractional Kelly) with guardrails

6. Functional Requirements

Requirements are prioritized with MoSCoW (Must / Should / Could). "Must" items define the shippable core.

6.1 Live tournament experience

- **[Must]** Live and scheduled fixtures with model win/draw/loss probabilities on each upcoming match.
- **[Must]** Team pages with full squad, head coach, group standing, power rating and forecast.
- **[Must]** Player pages with per-90 stats and a percentile radar versus positional peers.
- **[Should]** Match detail with lineups, shot map and a plain-language match summary.

6.2 Forecasting

- **[Must]** Championship probability, stage-reach probabilities (R16→Final) and group-win odds, all from Monte Carlo simulation.
- **[Must]** A projected knockout bracket built from current qualification probabilities.
- **[Should]** Golden Boot projection that regresses current goals toward expected goals (xG).

6.3 Multi-tournament & discovery

- **[Must]** A single control that re-points the entire app between the live 2026 tournament and historical editions.
- **[Should]** Club Connections: every player matched to their club, navigable by league, club and nation.
- **[Could]** A 3D globe to browse nations spatially — spin, zoom, click a country to open its squad and history.

6.4 Decision support

- **[Should]** Betting Edge: model probability vs de-vigged market price, with EV flags and fractional Kelly staking, behind a non-dismissable responsible-gambling notice.
- **[Could]** Comparison cards: stack any players or teams side-by-side on chosen metrics and export a shareable card.
- **[Should]** Natural-language "Ask" — type a plain-English question and get a data-grounded answer.

7. User Experience Principles

1. **Live-first.** Default to the in-progress tournament; never make the user choose a dataset before seeing value.

2. **Explainable by default.** Every number is one hover or one link from a plain-language definition (see the in-app Guide).
3. **Dense but legible.** A terminal-grade information density, kept readable through a disciplined type and color system.
4. **Never a dead end.** Under partial or still-loading data, surfaces degrade to a sensible empty state — they never crash or show a blank screen.
5. **Responsible by construction.** Anything touching gambling leads with risk framing and treats a positive "edge" as a sign the model is probably wrong, not a tip.

8. Release Plan

Phase	Scope	Outcome
P0 — Engine	Domain model, simulation data, analytics engine, core pages.	Fully interactive product on synthetic data.
P1 — Real history	StatsBomb adapter; 2018/2022 men's & 2019/2023 women's tournaments.	Real, explorable historical analytics.
P2 — Live	API-Football adapter; live 2026 squads, fixtures, standings.	Live-first experience.
P3 — Depth	Club Connections, Discoveries, Betting Edge, 3D globe, Ask.	Differentiated, full-feature product.

9. Risks & Mitigations

Risk	Impact	Mitigation
Live feed rate-limits during burst load	Missing squads / partial data	Retry-with-backoff, paced batching, graceful empty states
Historical data gaps (open-data coverage)	A tournament missing a team/metric	Per-surface degradation; never block a page on one missing entity
Model mis-read as certainty	User over-trusts a probability	Always state method & sample size; expose full distribution, not a point pick
Responsible-gambling concerns	Reputational / ethical risk	Non-dismissable framing; no tips; illustrative-only on non-live data

10. Open Questions & Future Work

- Add women's 2027 and continental tournaments (Euros, Copa) as further switchable datasets.
- Optional accounts to sync favorites and saved comparison cards across devices.

- Richer AI narration (full match reports) gated behind a hosted LLM, with the deterministic generator as the always-on fallback.

Appendix A — Glossary

Term	Plain-language meaning
xG (expected goals)	The quality of a chance — how likely an average shot from that spot is to score.
ELO	A single strength rating; the gap between two teams' ELO implies a win probability.
Monte Carlo	Playing the rest of the tournament thousands of times and counting outcomes to get probabilities.
De-vig	Stripping the bookmaker's margin out of odds to recover the market's true implied probability.
Kelly stake	A formula for bet sizing given an edge; the product shows a conservative fraction, for illustration only.